Evaluating Agent Swarms: Token Budgets, Active Workers, and Baseline Choice

Maximilian Nicholson[1,2] Ayda Yazdani[1]
[1] University of Bath [2] Amazon
{mn866, ay582}@bath.ac.uk

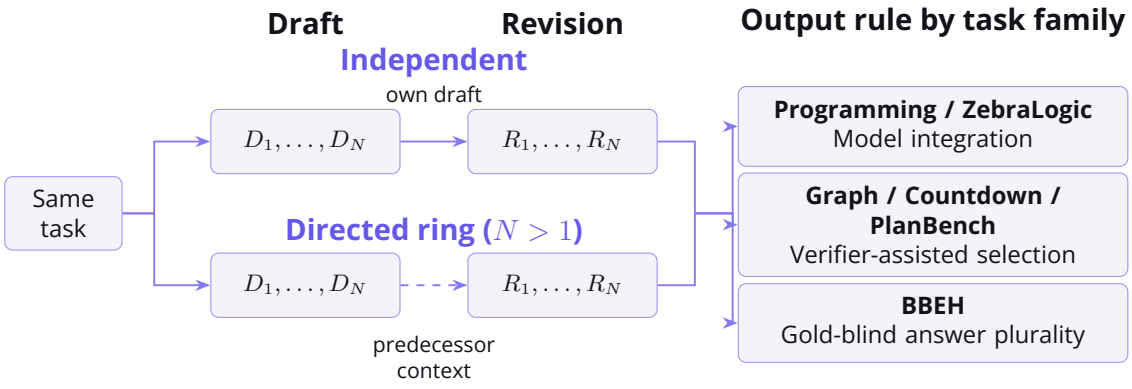
Abstract

Adding workers changes token use, worker participation and answer selection as well as coordination. We examine eleven scaling studies with 20,376 evaluation trials across seven task families and 6,272 calibration trials, using Luna and Muse Spark at requested widths up to sixteen. All 148 wider configurations use more recorded tokens than their matched-ceiling single-worker controls: 75 score lower, 24 tie and 49 improve. Budget-dependent feasibility is visible on PlanBench: independent four-worker accuracy rises from 0% to 100% when the ceiling doubles, while truncated stages fall from 99.94% to 0.19%. On BigCodeBench, Muse's sixteen-worker activation ranges from 7.3% to 93.5% across three ceilings. A fresh 300-task ZebraLogic confirmation transfers a communicating 4/8/16 policy without refitting, but its advantage over fixed four remains uncertain (+0.92 percentage points; 95% CI $[-0.58, 2.50]$). Ring revision provides no consistent advantage over independent revision. These heterogeneous studies use task-paired analyses, study-specific output rules and amended operational protocols. Recorded token debit does not measure exact total compute or retry expenditure. The findings motivate evaluating allocation policies with execution records and competitive controls; equal realized compute advantages and general worker-scaling laws remain unestablished.

Keywords: Agent evaluation; Multi-agent systems; Test-time compute; Token budgets; Resource allocation; Coordination; Reliability; Reproducibility

"All happy families are alike; each unhappy family is unhappy in its own way." —Leo Tolstoy, *Anna Karenina* (Tolstoy, 2014).

Adapted to this study: All successful swarms share a valid outcome; unsuccessful swarms fail in different ways.



Nominal stage shares: programming / ZebraLogic: 50% draft, 25% revision, 25% integration; later graph / Countdown / BBEH / PlanBench diagnostics: 50% draft, 50% revision.

Figure 1 | Study-specific execution. Independent revision uses its own draft; ring revision receives predecessor context. Both conditions share the study's output rule. Requested width N can exceed active participation when prompt reserves prevent calls. Tested widths are 1/2/4/8/16 where applicable; nominal ceilings and retained token debit are distinct.

1 Design and Findings

We audit worker width as a resource-allocation decision (Snell et al., 2024; Kim et al., 2026).

Design. We analyse subsets of BigCodeBench, MBPP/EvalPlus, ZebraLogic, BBEH and PlanBench (Zhuo et al., 2025; Austin et al., 2021; Liu et al., 2023; Lin et al., 2025; Kazemi et al., 2025; Valmeekam et al., 2023), plus custom graph/Countdown tasks with deterministic checks; graph generation is inspired by Reasoning Gym (Stojanovski et al., 2025). Widths reach sixteen in coding and ZebraLogic; later diagnostics test one and four. Model versions and routing vary. Figure 1 distinguishes output rules; verifier-assisted gains include selection.

Budget rules. PlanBench's independent $N = 4$ effect changes from -95.0 pp (CI $[-98.0, -91.5]$) to $+10.5$ pp (CI $[6.0, 15.5]$) as its ceiling doubles. Higher-ceiling debit is 4.06 times the single-worker control; stage truncation falls from 99.94% to 0.19%. Figure 2 summarizes 148 wider cells sharing tasks and controls.

Worker participation. BigCodeBench Muse's $N = 16$ activation is 7.3%, 45.3% and 93.5% at ceilings 32,768, 65,536 and 131,072. At 32,768, independent widths 1/8/16 pass 40/64, 36/64 and 14/64. Activation's causal role remains unresolved.

Baseline choice. On 300 fresh ZebraLogic puzzles, transferred ring $4/8/16$ minus fixed four averages $+0.92$ pp at 65,536/131,072 tokens ($[-0.58, 2.50]$), establishing neither superiority nor equivalence. At 131,072, sixteen communicating workers score 90.33% versus four's 89.67%, using 3.57 times retained debit.

Coordination. Sampling, aggregation and debate are distinct mechanisms (Brown et al., 2024; Wang et al., 2023; Du et al., 2023). Countdown's independent gains of 9.5–17.0 pp use 2.77–3.85 times recorded tokens; its pooled ring effect is -5.0 pp ($[-8.5, -1.75]$). Seven Muse graph zeros are operational failures, not necessarily reasoning errors. These ring results have narrower scope than shared-workspace communication (Park et al., 2026); execution audits complement failure taxonomies (Cemri et al., 2025).

Uncertainty and limits. Task bootstrap retains configurations and repetitions, stratified where applicable (Efron, 1979; Field and Welsh, 2007); reported 95% CIs lack cross-study multiplicity correction. Pilots and smoke checks are excluded. Retained debit omits retry expenditure and can include conservative charges for unknown usage; equal realized compute is not established.

These cases motivate three checks for scaling claims: execution feasibility, realized worker participation and competitive fixed-width baselines. Ongoing work tests policy transfer across resource divisions.
https://github.com/PluvioXO/agent-swarms-reproducibility.

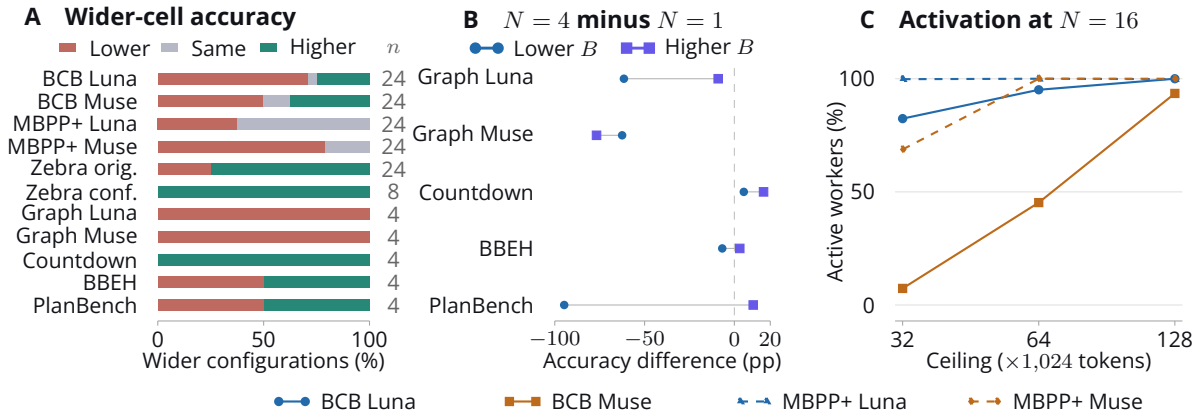


Figure 2 | Evaluation results. (A) Wider cells versus matched-ceiling $N = 1$; n counts configurations. (B) Four-worker effects pooled over revision conditions, at ceilings 4/8k for graphs and 8/16k otherwise ($k = 1{,}024$ tokens). (C) Activation at $N = 16$.

References

Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, et al. Program synthesis with large language models, 2021. https://arxiv.org/abs/2108.07732. Cited at: 1.

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V. Le, Christopher Ré, et al. Large language monkeys: Scaling inference compute with repeated sampling, 2024. https://arxiv.org/abs/2407.21787. Cited at: 1.

Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, et al. Why do multi-agent LLM systems fail?, 2025. https://arxiv.org/abs/2503.13657. Cited at: 1.

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate, 2023. https://arxiv.org/abs/2305.14325. Cited at: 1.

Bradley Efron. 1979. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1): 1–26. https://doi.org/10.1214/aos/1176344552. Cited at: 1.

C. A. Field and A. H. Welsh. 2007. Bootstrapping clustered data. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(3):369–390. https://doi.org/10.1111/j.1467-9868.2007.00593.x. Cited at: 1.

Mehran Kazemi, Bahare Fatemi, Hritik Bansal, John Palowitch, Chrysovalantis Anastasiou, Sanket Vaibhav Mehta, et al. BIG-Bench extra hard, 2025. https://arxiv.org/abs/2502.19187. Cited at: 1.

Yubin Kim, Ken Gu, Chanwoo Park, Chunjong Park, Samuel Schmidgall, A. Ali Heydari, et al. Towards a science of scaling agent systems, 2026. https://arxiv.org/abs/2512.08296v3. Cited at: 1.

Bill Yuchen Lin, Ronan Le Bras, Kyle Richardson, Ashish Sabharwal, Radha Poovendran, Peter Clark, et al. ZebraLogic: On the scaling limits of LLMs for logical reasoning, 2025. https://arxiv.org/abs/2502.01100. Cited at: 1.

Jiawei Liu, Chunqiu Steven Xia, Yuyao Wang, and Lingming Zhang. Is your code generated by ChatGPT really correct? Rigorous evaluation of large language models for code generation, 2023. https://arxiv.org/abs/2305.01210. Cited at: 1.

Jongho Park, Vasilis Kontonis, Shivam Garg, Akshay Krishnamurthy, and Dimitris Papailiopoulos. Scaling discovery through test-time communication, 2026. https://arxiv.org/abs/2609.21032. Cited at: 1.

Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling model parameters, 2024. https://arxiv.org/abs/2408.03314. Cited at: 1.

Zafir Stojanovski, Oliver Stanley, Joe Sharratt, Richard Jones, Abdulhakeem Adefioye, Jean Kaddour, et al. Reasoning Gym: Reasoning environments for reinforcement learning with verifiable rewards, 2025. https://arxiv.org/abs/2505.24760. Cited at: 1.

Leo Tolstoy. *Anna Karenina*. Penguin Classics, 2014. ISBN 9780141199610. https://www.penguinrandomhouse.com/books/530530/anna-karenina-by-leo-tolstoy/9780141199610/. Translated by Richard Pevear and Larissa Volokhonsky; Part One, Chapter 1. Cited at: 1.

Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change, 2023. https://arxiv.org/abs/2206.10498. Cited at: 1.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, et al. Self-consistency improves chain of thought reasoning in language models, 2023. https://arxiv.org/abs/2203.11171. Cited at: 1.

Terry Yue Zhuo, Minh Chien Vu, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, et al. BigCodeBench: Benchmarking code generation with diverse function calls and complex instructions, 2025. https://arxiv.org/abs/2406.15877. Cited at: 1.